



Identifying novel customer needs from user-generated content for product development using pre-trained language model

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ABSTRACT

The identification of novel customer needs is crucial for companies to create new products and seize business opportunities in a constantly evolving technological and social landscape. However, traditional methods for identifying emerging needs are costly, time-consuming, and labour-intensive, often resulting in delays in bringing products to market. In recent years, user-generated content (UGC), such as online product reviews, has emerged as a promising alternative source for uncovering novel customer needs. In this paper, we propose a novel approach to identifying customer needs by treating this as a text classification task. Specifically, we leverage the power of the pre-trained language model BERT to analyze and extract insights from UGC, particularly online product reviews. To address the challenge of class imbalance in the data, we developed a regularized dual BERT structure that achieves state-of-the-art performance. Our experiments demonstrate the effectiveness of this structure, showing that it is robust even when dealing with reviews of varying lengths. By using this proposed methodology, companies can quickly and efficiently automate the process of identifying novel customer needs, requiring fewer expert resources.

ARTICLE HISTORY

Received 1 September 2024
Accepted 8 May 2025

KEYWORDS

BERT; customer needs; language model; UGC

Introduction

Customer needs have long been considered the driving force for companies to identify new product opportunities, to enhance the design of new products and to improve existing products (Ulrich and Eppinger 2006). Socio-economic changes and rapid technological innovation have led to the evolution of existing customer needs and the emergence of new needs (Luo 2023). This provides opportunities for companies to develop new products to meet these needs and turn them into profit-making ventures. For example, sport utility vehicles (SUVs) were developed in 1980s to meet the novel customer need for a luxury vehicle that could haul moderate loads. Today, the SUV is one of the most profitable segments

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of the automotive industry (Urban and Hauser 2004). Novel customer needs can also benefit product development for well-established, mature products. For example, to capture the Asian trend to use smaller, flat packaging, Kao Corporation successfully introduced a highly concentrated detergent, Attack, to meet customer needs, offering small-package detergents that, nevertheless, had the same cleaning power (Timoshenko and Hauser 2019). Kao gained the competitive advantage by being the first to identify and fulfill a newly emerging need.

Product development and marketing teams across a wide range of industries are continually seeking ways to understand customers and identify novel needs for products opportunities. Typical tools used by industries include marketing surveys, focus groups, and the lead user method. However, these traditional methods are expensive, and labour – and domain knowledge-intensive. For example, General Motors (GM) used to spend tens of millions of dollars annually searching for newly emerging needs (Urban and Hauser 2004). In addition, traditional methods are time-consuming and less effective amidst the high level of global competition. Currently, product developers face fast time-to-market pressure from their competitors and customers, and as such, there is a growing demand for a fast, efficient, and accurate technique for detecting novel needs at the early product design stage.

Nowadays, the internet is full of user-generated content (UGC), which comprises rich textual data such as online product reviews (Dong et al. 2024). UGC is continuously accumulated and updated, and can be rapidly and inexpensively extracted. Therefore, it provides a promising data source for more efficiently identifying novel customer needs. The analysis of UGC has garnered significant attention from researchers seeking to understand consumer needs. Online reviews, in particular, provide a wealth of implicit information that reflects customer preferences across various aspects of products or services (Wang et al. 2024). Recent research has acknowledged that mining the UGC corpus can identify customer needs with a performance at least comparable to that of traditional methods (Timoshenko and Hauser 2019). However, there is a dearth of efficient techniques for identifying novel customer needs from UGC text. Many companies still manually screen UGC data to generate new product design ideas. For example, C&A Marketing uses an entire team of professionals to monitor reviews on eBay and Amazon to identify novel customer needs for new product development (Lee, Manzoor, and Cheng 2019).

To develop more efficient ways of mining novel needs from the UGC corpus, some critical challenges must be overcome. Firstly, UGC data are unstructured and mostly in free-form text. They are not particularly generated by customers to express their novel needs. It is necessary to comprehend the rich meaning underlying the constituent words using advanced natural language processing technologies. Secondly, the emergence of novel needs is usually unexpected. They are not articulated in the existing UGC corpus, as very few customers have expressed these needs previously. Furthermore, only a small portion of reviews contain novel need information. To ensure that new product development opportunities are not overlooked, a company's need-detection methods should be sensitive enough to identify novel needs as soon as they arise. Finally, UGC contains a large amount of irrelevant information to the product, such as the complaints about the sales and service. This further complicates the task of extracting the desired information from the large amount of data. An effective method must filter out irrelevant information and focus on the informative content.

This study aims to address the aforementioned challenges by proposing an efficient method for mining UGC to identify novel customer needs. To tackle this problem, we approach it from the perspective of text classification, through which we classify reviews as either positive (containing novel customer needs) or negative (lacking novel content). To gain a comprehensive understanding of the complex language used in unstructured reviews, we extract semantic features using the powerful pretrained language model, BERT. However, the class imbalance issue is significant, as our dataset exhibits a highly skewed ratio of positive to negative samples (1:88). To overcome this issue and improve the effectiveness and efficiency of identifying novel need content, we propose a regularized dual BERT structure. This approach leverages the concept of Kullback-Leibler (KL) divergence and incorporates it into the model. By introducing KL loss, the regularized BERT framework encourages both BERT networks to generate probability distributions that are more aligned with each other, fostering a unified understanding of the input data. This regularisation mechanism helps prevent overfitting to the majority class and ensures a more balanced representation of both positive and negative instances. Our experiments demonstrate the effectiveness of the regularized BERT structure in accurately identifying novel customer needs. By mitigating the class imbalance issue and promoting a more balanced representation, this approach proves to be a valuable tool in the identification process.

This paper contributes in several ways. Firstly, we reframe the task of identifying novel customer needs as a text classification problem. To support our research, we have curated and annotated a dataset consisting of 16,667 laptop reviews. Secondly, we propose a regularized BERT structure to address the challenge of class imbalance. By employing dual BERT units, we introduce differential encoding that acts as a form of implicit data augmentation. This augmentation introduces variability in the learned representations during training, enhancing the understanding of the minority class. This approach facilitates improved generalisation and recognition of novel patterns associated with customer needs expressed in reviews. Lastly, our experiments demonstrate that the proposed method achieves state-of-the-art performance. We showcase the effectiveness of the regularisation mechanism and highlight the robustness of our approach when dealing with reviews of varying lengths. These findings validate the efficacy of our method in accurately identifying novel customer needs. Overall, this research contributes to the field by reframing the problem, developing a novel regularisation technique, and achieving impressive performance results. The release of the dataset and code further enhances the reproducibility and extendibility of this research.

Relevant work

Traditional methods for identifying customer needs

For decades, identifying customer needs has been a key research area in product design and marketing. Traditional methods often involve face-to-face interactions or similar settings during the initial design phase. These methods include focus groups, the lead user method, quality function deployment (QFD), and conjoint analysis. Focus groups facilitate group discussions, allowing customers to share their perceptions, opinions, and preferences about products (Morgan 1997). Companies believe that focus groups can uncover a wider range of customer needs because participants build on each other's ideas, creating a cascading

effect. However, some studies suggest that focus groups may suffer from opinion distortion and selective bias to reach predetermined conclusions (Rushkoff 2005).

The lead user method is also employed in the industry for product innovation. Lead users are those who experience needs that are not yet recognised by the general public. Companies seek out and interview these lead users to discover new product design ideas. However, this method can be time-consuming, and identifying lead users is often challenging (Cooper and Edgett 2009). QFD involves a series of mappings to extract customer needs and translate them into technical specifications for product design and production (Hunter 1994). A typical QFD process requires the participation of multiple stakeholders, including customers, designers/engineers, and marketing teams. It relies on customers' ability to articulate their needs and the designers' and engineers' expertise to interpret these needs. The entire process can be labour-intensive and time-consuming (Martins and Aspinwall 2001).

Conjoint analysis asks customers to make trade-offs by evaluating combinations of multiple predefined product attributes (Green and Rao 1971). Statistical methods are then used to determine how customers value different attributes. However, conjoint analysis confines customer needs to a fixed design space, limiting the expression of general needs. In summary, traditional methods face challenges related to efficacy (such as conjoint analysis) or efficiency (such as focus groups and QFD) in identifying new customer needs.

Text mining for customer needs analysis

The impact of online customer reviews on customer-centric product innovation has been extensively studied (Zhou et al. 2018). For instance, Zhang, Rao, and Feng (2018) demonstrated strong correlations between changes in smartphone features and online reviews. Topics relevant to this proposal include customer needs identification and analysis, and opinion mining. This subsection summarises some representative research work in this area.

Researchers have explored methods to extract opinions from product reviews, including linguistic rule-based techniques (Jindal, Liu, and Lim 2010), sequence labelling (Huang et al. 2012), and topic modelling (Büschken and Allenby 2016). Among these, Latent Dirichlet Allocation (LDA) is widely used, clustering words into topics with Dirichlet priors (Blei, Ng, and Jordan 2003) and extracting product-feature topics (Brody and Elhadad 2010). Jo and Oh (2011) enhanced LDA with opinion categorisation to identify product features and opinions simultaneously. MaxEnt-LDA combines maximum entropy with LDA to extract aspects and related opinions (Zhao et al. 2010). Zhou et al. (2020) used LDA for customer needs topics and applied sentiment analysis to gauge sentiment intensity. Zhou et al. (2024) linked product reviews with purchasing records using multi-view LDA to deduce customer preferences and connect topics to purchasing motivations.

In product design research, online reviews have also been leveraged to identify customer preferences. Zhai et al. (2011) summarised review texts to rank important design needs. Liu, Lu, and Loh (2007) combined domain knowledge with reviews via a knowledge management system. Liu and Toubia (2018) introduced a dual LDA model using search queries and web pages to infer preferences. Decker and Trusov (2020) aggregated needs from text to evaluate product features' impact on satisfaction. Wang et al. (2011) developed a design method based on extracted product attributes and a hierarchical preference model. Liu

et al. (2013) employed a regression model to assess review helpfulness for boosting product satisfaction. Recent studies have focused on using online reviews to extract interpretable factors for business. For example, Chen et al. (2019) used reviews to capture textual features as weak supervision signals to enhance visual preference learning, showing improved recommendation performance and interpretability. Liu et al. (2019) proposed a dual attention mutual learning algorithm to jointly extract features from reviews and interactions, using them to represent user preferences and enhance recommendation interpretability. Timoshenko and Hauser (2019) utilised a convolutional neural network to pinpoint and group informative sentences about products from review texts. Analysts then manually examined these sentences to uncover customer needs. However, this approach was not tailored for identifying novel customer needs and relied on human expert input rather than an automated process.

Some research has focused on extracting defect information from online reviews for product development. Abrahams et al. (2012) used logistic regression to detect vehicle defects from online forums and later proposed a text analytics framework for uncovering product defects in user-generated content (Abrahams et al. 2015). Qiao et al. (2017) developed an LDA model to gather detailed information about product defects from online reviews. However, it's important to note that defects found in reviews differ from novel customer needs. To our knowledge, no research has yet focused on automatically identifying novel needs from user-generated content. This area of research is still emerging, with few studies dedicated to developing advanced machine learning methods for automatically extracting product innovation ideas from online reviews.

Methods

Problem definition

The basic notations to be used in this research are defined as follows. Let $\mathbf{R}$ be the review corpus of a product, such as laptops. Let vector $\mathbf{x} \in \mathbf{R}$ be a review in the corpus. Each review $\mathbf{x}$ is associated with a binary label y with value 1 indicating there is novel need content in the review, and 0 indicating otherwise. Given the training set $\{\langle \mathbf{x}_i, y_i \rangle\}_{i=1}^{|\mathbf{R}|}$, where $\langle \mathbf{x}_i, y_i \rangle \in \mathbf{R} \times \mathbf{Y}$ and $|\mathbf{R}|$ is the cardinal of set $\mathbf{R}$, i.e. the number of elements in $\mathbf{R}$, and $\mathbf{Y}$ is the label space, we aim to develop classifier f to identify reviews containing novel need content, $f: \mathbf{R} \rightarrow \mathbf{Y}$. Then, given a new review $\mathbf{x}^*$, we can tell whether the review contains novel need content by applying $f(\mathbf{x}^*)$.

The regularized dual BERT-based structure

Users generate reviews in free-form text, necessitating state-of-the-art natural language processing to understand the underlying semantics. To overcome the difficulties presented by informal expressions in reviews and the imbalance of classes in the dataset, we deployed a regularized dual BERT-based structure to identify the positive reviews, which contain novel need content, as shown in Figure 1. Since its emergence, the pretrained language model BERT and its variants have achieved state-of-the-art performance in many natural language processing tasks (Devlin et al. 2019; Luo et al. 2023; Li et al. 2025). However, because the neural network structure is highly susceptible to overfitting during training,

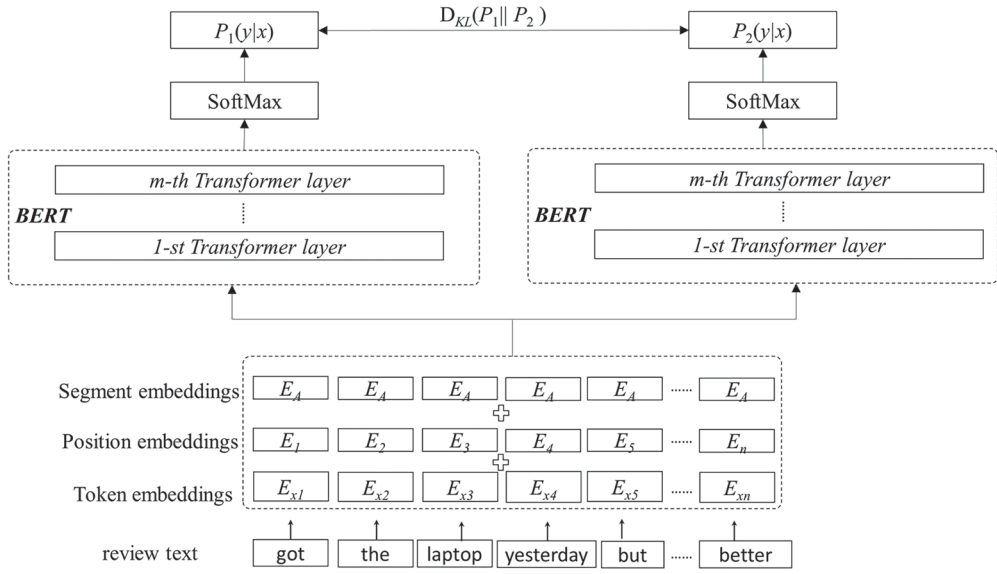


Figure 1. The dual BERT-based structure used to identify positive reviews.

the dropout operations were added to the BERT units to prevent overfitting of the model, meaning that the values of certain dimensions of text encoding were randomly set to be 0. Although that operation overcame model overfitting, it randomised the trained neural network as well, which led to non-robust results regarding positive review identification. To mitigate the randomness induced by the random dropout operations, we used a dual BERT-based structure to regularise the results of training. In particular, the text was input into two identical BERT networks as shown in Figure 1. After passing the SoftMax layer, two different probability distributions on the label sets were generated. We used Kullback – Leibler (KL) loss to regularise the random dropout operations in those two BERT networks and to stabilise the final identification of positive reviews (MacKay 2003). When training the network, we combined both the cross-entropy losses in the BERT units as well as the KL loss between the two probability distributions. The dropout layers in the two BERT units produce different encodings of the same review. Thus, the dual BERT network served as a kind of data augmentation, and thus, could somewhat alleviate the imbalance of categories in the training set.

BERT as the embedding layer

We used BERT to encode the text in reviews given its superior performance in automatically extracting the semantic features from text.

Let us consider a review $\mathbf{x} = (x_1, x_2, \dots, x_n)$ consisting of n words, with its corresponding class label denoted as y . For each word x_i , we represent the word embedding, position embedding, and segment embedding as $E(x_i)$, $E(i)$, and $E(l_x)$, respectively (Devlin et al. 2019). These embeddings are then combined using the concatenation operation, resulting in the combined embedding $h_i = E(x_i) + E(i) + E(l_x)$. The whole reviews can thus be encoded as $\mathbf{h} = (h_1, h_2, \dots, h_n)$. Subsequently, $\mathbf{h}$ runs through a normalisation layer and dropout layer and is fed into BERT such that $\mathbf{h}^0 = \text{Dropout}(\text{LN}(\mathbf{h}))$, in which Dropout and

LN represent a dropout operation and layer normalisation, respectively. Layer normalisation computes the mean and standard deviation of all the reviews' encoding across the training examples, and uses these statistics to normalise $\mathbf{h}$. LN can improve the efficiency and stability of the model. The dropout layer is randomly set such that certain dimensions' value of $LN(\mathbf{h})$ is 0 in order to avoid model overfitting. Dropout is a simple yet effective technique that has been shown to improve the performance of neural networks in various tasks (Vaswani et al. 2017). Due to its randomness, the outputs of the dropout layer is not deterministic.

Our BERT network consists of multilayered transformer blocks. For transformers in layer $i + 1$, the output is calculated as $\mathbf{h}^{i+1} = Skip(FF(\mathbf{h}^i), Skip(MHSA, \mathbf{h}^i))$, in which MHSA represents the multiple head self-attention function, $Skip(f, \mathbf{h}) = LN(\mathbf{h} + Dropout(f(\mathbf{h})))$, and $FF(\mathbf{h}) = GELU(\mathbf{h}\mathbf{W}_1^T + \mathbf{b}_1)\mathbf{W}_2^T + \mathbf{b}_2$ (Devlin et al. 2019). GELU is Gaussian Error Linear Unit which represents the element-wise activation function (Hendrycks and Gimpel 2016), and $\mathbf{W}_1, \mathbf{W}_2, \mathbf{b}_1, \mathbf{b}_2$ are the parameters of weight matrices and bias vectors to be estimated during training. After being submitted to m layers of transformer blocks, the reviews can be encoded as the feature $\mathbf{h}^m = (h_1^m, h_2^m, \dots, h_n^m)$, which is represented by $\mathbf{H}$ in this paper.

In the model structure illustrated in Figure 1, the reviews are encoded and input into two dual BERT units such that two encodings can be obtained. We let $\mathbf{H}_1$ and $\mathbf{H}_2$ be the output of the two BERT units given the same input review x . Because the BERT network consists of multiple layers of transformer blocks and thus multiple random dropout layers, the output of those two BERT units, $\mathbf{H}_1$ and $\mathbf{H}_2$, are different encodings of the same content. As a result, the dual BERT-based structure augments the original dataset, and data sparsity can be somewhat alleviated.

Label distribution derivation via the softmax layer

The BERT encoding $\mathbf{H}$ is fed into a SoftMax layer to derive the probability distribution that needs $\mathbf{x}$ belongs to y in the label set (Wang et al. 2021). To be specific, $P(y = y_k | \mathbf{x}) = \frac{\exp(\mathbf{W}_k \cdot \mathbf{H})}{\sum_j \exp(\mathbf{W}_j \cdot \mathbf{H})}$, in which y_k is a binary label indicating whether the need is novel or not. The parameter $\mathbf{W}_k$ can be estimated by minimising the cross-entropy loss (LCE) between the predicted label (i.e. the label with the highest probability) and the true label distribution, i.e. $L_{CE} = - \sum_{i=1}^{|\mathbf{T}|} \sum_{j=1}^{|\mathbf{C}|} y_j^i \log P_j^i$, in which P_j^i is the predicted probability that the i -th review has label j using the SoftMax output and that y_j^i has a value of 1 when the i th review has label j . $|\mathbf{T}|$ and $|\mathbf{C}|$ are the size of the training dataset and the number of labels, respectively.

Using a dual BERT-based structure, two encodings of the same review x , namely $\mathbf{H}_1$ and $\mathbf{H}_2$, are fed into two independent SoftMax layers to derive the probability distribution that the needs x will be assigned a label y . As a result, we have two cross-entropy loss functions, $L_{CE}^{(1)}$ and $L_{CE}^{(2)}$, and two probability distribution functions, $P_1(y | \mathbf{x})$ and $P_2(y | \mathbf{x})$ where y is the label of x .

The regularisation of the dual BERT units

After being submitted to the dual BERT-based structure, the review x has two probability distributions over the possible label set, $P_1(y | \mathbf{x})$ and $P_2(y | \mathbf{x})$, which are simplified as P_1 and P_2 , respectively. Their KL divergence between P_1 and P_2 is defined as $D_{KL}(P_1 || P_2) = \sum P_1 \log \frac{P_1}{P_2}$ (MacKay 2003), which measures how the probability distribution of P_1 differs

from P_2 . Symmetric KL loss is used to quantify the difference between those two distributions – that is, $L_{KL}(P_1||P_2) = \frac{1}{2} \sum_{\mathbf{x}} (D_{KL}(P_1||P_2) + D_{KL}(P_2||P_1))$. To enhance the robustness of identifying novel needs and mitigating the impact of random dropout operations, we aim to minimise the KL loss between the distributions during the model training stage.

We combined the cross-entropy loss in each BERT unit and the KL loss, which regularises the dropout operations as $L = L_{CE}^{(1)} + L_{CE}^{(2)} + \lambda L_{KL}(P_1||P_2)$. λ is a parameter to control the relative weight of KL loss. During model training, we estimated all parameters in the dual BERT network by minimising that total loss.

Experiment setting

Data set

We crawled reviews of laptops introduced from 2017 to 2022 from Amazon.com., and in total, found 16,667 reviews. All reviews were manually annotated by two annotators with IT backgrounds, independently using the label positive (for reviews containing novel need content) and negative (for reviews without novel need content). Fleiss's kappa statistic is commonly employed to assess the level of agreement between annotators in inter-annotator agreement evaluations. In this study, the obtained kappa value was 0.91, suggesting a substantial level of agreement among the annotators (Fleiss, Levin, and Paik 2003). In cases where the reviews were annotated with differing labels, two annotators and an experienced consolidator in the field of IT engaged in additional discussions to resolve disagreements in the annotations. Among them only 187 were positive reviews, indicating a significant class imbalance issue, with the ratio around 1:88 between positive and negative samples. Some positive samples are shown below with novel needs content highlighted with bold.

- This laptop is fast and very responsive though it takes awhile getting it there so make sure you go through the Windows 10 settings with an fine toothed comb. **it would've been great if it came with the Ethernet USB dongle and the USB type C charger.**
- It would be fantastic if the laptop **has a projector.**
- The computer's appearance is very exquisite, and the metallic texture is excellent. I suggest **adding a waterproof function, with safe waterproofing** within 10 meters, so you don't have to worry about the computer getting wet in extreme environments, at least not worrying about the situation where a water cup next to the computer spills.

To ensure that the model was adequately trained and reliably evaluated, we randomly divided the corpus into training, validation, and testing sets in proportions of 40%, 20%, and 40%. These were used to train the model, select the optimal parameters, and test the models' performance.

Performance measure

Since this was a binary text classification problem, we mainly used precision, recall, and F1 scores, the popular evaluation metrics used in classification tasks, to evaluate the

performance (Lyu et al. 2024). Precision measures the proportion of correctly predicted positive samples out of all samples predicted as positive. A high precision value implies that the model has a low rate of false positives, which is crucial in scenarios where misclassifications can have significant consequences. Recall measures the proportion of correctly predicted positive samples out of all actual positive samples. It represents the model's ability to identify all positive instances without missing any. The F1 value is the harmonic mean of precision and recall (Huang et al. 2024). It provides a single metric that balances both precision and recall, considering both false positives and false negatives. Mathematically, $P = \frac{TP}{TP+FP}$ and $R = \frac{TP}{TP+FN}$ where TP (FP) is the number of correctly (incorrectly) classified positive samples. FN is the number of samples that should be classified as positive but incorrectly classified as negative, i.e. false negative. $F1 = \frac{2*PR}{P+R}$.

It should be noted that when monitoring online reviews to identify novel customer needs, only a very small proportion of the reviews contain novel need content. Thus, it would be quite time consuming and labour intensive for designers or relevant experts to screen all the reviews and identify the reviews containing novel needs.

We used the developed model to rank all the reviews based on the likelihood of containing novel need content. Since the designers only needed to screen the reviews with a high likelihood of novel need content, the efficiency of identifying these was greatly improved. To be specific, the trained model was tested using the testing set containing both positive (i.e. the reviews containing novel customer needs) and negative samples (i.e. the reviews without novel customer needs content). Each testing data went through the trained model and obtained a novelty score based on the SoftMax layer's output. Then all the reviews were sorted based on the novelty score, indicating the degree of novelty content in the reviews. In such a scenario, the performance metrics of precision, recall, and F1 in the top-k ranked samples, $P@k$, $R@k$, and $F1@k$, which are widely used in information retrieval research, were used as the supplemental evaluation criteria. Rigorously, $P@k = \frac{TP@k}{TP@k+FP@k}$ and $R@k = \frac{TP@k}{TP@k+FN@k}$ where $TP@k$ ($FP@k$) represent the number of correctly (incorrectly) classified positive samples among the top-k ranked samples. $FN@k$ is the number of samples that should have been classified as positive but were incorrectly classified as negative among the top-k ranked samples.

Experiment results

The effect of regulation parameters on the performance

The training batch size was set at 32. We set the hyperparameters (e.g. learning rate) by trial in relation to performance. For example, the learning rate of the Adam optimiser was set to be $2e-5$, as it slightly outperformed other settings. We used Nvidia Tesla V-100 GPU to train the model and PyTorch to implement the method.

In the model training stage, the total loss function is the sum of the cross-entropy loss of the two BERT units and the KL loss, which regulates the two BERT units. The KL loss term in the total loss function serves as a regularisation component. Regularisation techniques aim to prevent overfitting and enhance the model's generalisation ability. Achieving an optimal balance between regularisation and model flexibility is crucial. The identification of novel customer needs requires a high level of adaptability to diverse expressions within reviews. Allowing a certain level of divergence between the two BERT networks through a smaller KL

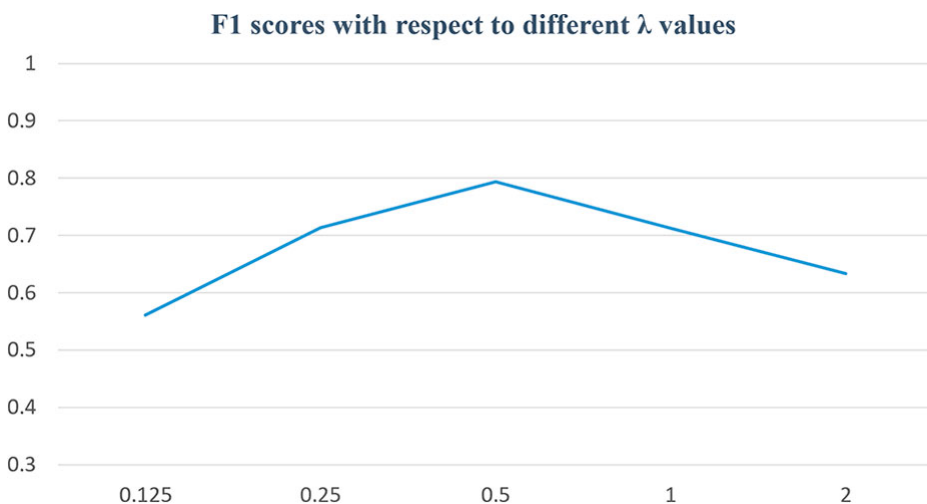


Figure 2. Overall F1 scores with respect to different λ values.

loss may encourage the model to be more task-specific and nuanced in its understanding of the imbalanced data.

We conducted experiments to select the suitable λ , which control the relative weight of KL loss with cross-entropy loss. The F1 values with different λ values appear in Figure 2. The figure shows that as λ increases, the F1 score initially rises before falling. When λ was approximately 0.25, the F1 value peaked. In our experiment, we chose $\lambda = 0.25$ because of the performance of the identification of novel needs. Thus, a relatively smaller KL loss is preferred.

While regularisation helps prevent overfitting and enhances generalisation, too much regularisation can hinder the model's capacity to capture important patterns. The smaller KL loss in this case strikes a balance, allowing the model to benefit from the regularisation without overly constraining its learning process. A smaller KL loss indicates a milder regularisation effect, allowing the model to retain more flexibility during training. In the context of identifying novel customer needs, a more flexible model may be advantageous. Smaller KL loss implies that the two BERT networks are allowed to diverge slightly in their predictions, introducing diversity into their learned representations. This diversity can be beneficial when dealing with the nuanced and varied expressions of novel needs in product reviews. If the KL loss is too large, it could force the two BERT networks to converge too closely, potentially over-regularizing the model. Over-regularisation might lead to a loss of crucial information and sensitivity, particularly in the presence of a highly imbalanced dataset and when trying to identify subtle patterns in negative reviews.

Overall performance

The regularized dual BERT structure achieved high overall precision, recall, and F1 performance, with $P = 0.9583$, $R = 0.8796$, and $F1 = 0.9152$. The performance evaluated among top-k samples can be found in Table 1. It can be seen that all of the top 25 reviews contain novel need content. The precision@k is 100% when k is smaller than or equal to 25. With

Table 1. The overall performance of the proposed method.

The value of k	Precision@k	Recall@k	F1@k
5	1.0000	0.0667	0.1250
10	1.0000	0.1333	0.2353
15	1.0000	0.2000	0.3333
25	1.0000	0.3333	0.5000
50	0.9600	0.6400	0.7680
75	0.8133	0.8133	0.8133
100	0.6400	0.8533	0.7314
150	0.4400	0.8800	0.5867
200	0.3300	0.8800	0.4800

the increase of k, some false negatives become included in the list, resulting in decreased precision and accuracy. However, the recall@k continues to increase as more reviews containing novel need content are identified and evaluated. When $k = 75$, which is the same as the number of positive samples (i.e. the reviews containing novel need content) in the testing data, the $\text{recall}@75 = 0.8133$. This indicates that 61 positive sample reviews were correctly identified from the testing set, $75 \times 0.8133 = 61$, which means only 14 positive samples were missing from the top 75 ranked reviews, thus showing the method's efficiency.

The testing data size was 6,667. Suppose a user randomly screened the testing review data. Then the user would be expected to screen $6667 \times 61/75 = 5422$ reviews to find the 61 positive samples. Using the proposed method, the efficiency can be improved by $5422/75 = 72$ times. With the increase of k, more positive reviews can be identified but at a diminishing speed. When $k = 150$, 66 positive reviews can be found.

Some false positive and false negative samples are shown below:

- False positive: This computer's robust build quality is truly impressive, it feels like it could survive a fall off a cliff. But the lack of a touchscreen is a real disappointment, it's like driving a car without power steering. I'm looking forward to the day when they'll introduce a responsive touchscreen that makes navigation a breeze.
- False positive: The cooling system is efficient, maintaining optimal performance during heavy-duty tasks. A silent operation mode for quieter environments would be a worthy addition.
- False negative: The old laptop at home broke down. Two colleagues at work use this laptop, the one who has been using it for the longest has been using it for over half a year. Both said that it's very good to use, high configuration, high cost performance, fast operation, and highly recommended this model. So, I also bought this model. After buying it, it didn't disappoint me. Full metal shell, fashionable appearance, fast operation, very fast boot up, it feels like it completes in less than ten seconds, the Office it comes with is just what I need, great! As a blockchain technology worker, I hope it can **integrate a distributed computer and then use the central computer to quickly and securely process and save data**. Finally, talking about customer service, very professional and patient. If you can't solve a problem, the customer service patiently replies one by one, teaches bit by bit, very good.

The two false positive samples both contain customers' expectations for the computer, such as touch screen and silent operation mode. However, these features are not novel enough for new product development, as other laptops already have these functions. On the other hand, the false negative sample is quite common in our experiment. The review is quite long, and although they contain novel needs content, it is diluted within the long irrelevant sentences. As a result, it becomes difficult for the method to detect these novel needs. A detailed discussion on the impact of text length on the performance can be found in a later section.

The effectiveness of the proposed regularized BERT model in the identification of novel customer needs in online product reviews is particularly noteworthy given the substantial class imbalance prevalent in the dataset (approximately 1:88). The model's success in mitigating the challenges posed by this significant imbalance can be attributed to its inherent capacity for nuanced learning and regularisation mechanisms. The two BERT units with random dropout function in the MHSA serve as a kind of data augmentation (Hinton et al. 2012). By employing dropout-based data augmentation and KL divergence regularisation, the model strategically tackles the imbalance issue. The dropout layers introduce variability in the representations, enabling the model to better discern intricate patterns within the minority class, which is crucial for capturing the often subtle expressions of novel needs present in negative reviews.

Simultaneously, the regularisation term, with a preference for smaller KL loss, fine-tunes the model's sensitivity without overly constraining its capacity to adapt to the imbalanced distribution (Wan et al. 2013). This dual approach not only enhances the model's robustness but also serves as a sophisticated countermeasure against the bias typically associated with majority class prevalence. The regularized BERT model, therefore, not only excels in handling class imbalance but leverages it as a pivotal aspect of its training strategy, ultimately leading to more accurate and balanced identification of novel customer needs.

Comparison with baseline methods

To demonstrate the effectiveness of our proposed method, we compare it with several popular text classification methods, including fastText, CNN, and BiLSTM. These methods are widely used in natural language processing (NLP) tasks, such as text classification. fastText, developed by Facebook, represents words as n-grams (subword units) and uses embeddings for these subword units to generate word embeddings (Joulin et al. 2016). This approach allows fastText to handle out-of-vocabulary words and capture word meanings based on their constituent character n-grams. CNN, on the other hand, utilises convolutions on the input word embeddings to extract local features. It then applies pooling to obtain a fixed-length representation of the text (Zhang et al. 2024), which is then fed into a fully connected layer for text classification. BiLSTM employs both forward and backward LSTM layers to encode the input text and generate a fixed-length representation (Wang, Zhao, and Wan 2021). This representation is then passed to a fully connected layer for classification. BiLSTM is particularly effective at capturing longer-range dependencies in the input text.

In dual CNN, we used 64-dimension GloVe embeddings and batch size 32 and set the kernel sizes to 2, 3, and 4. The learning rate was set to be $5e-3$ by trial, and the dropout rate was 0.5. In BiLSTM with attention, we used the same parameter settings for the dimensions of

Table 2. Comparison with popular baselines.

Model	Precision	Recall	F1
fastText	0.9394	0.5533	0.5927
CNN	0.9951	0.5667	0.6152
BiLSTM	0.8750	0.6661	0.7293
Regularized BERT	0.9583	0.8796	0.9152

embeddings, batch size, dropout rate, and learning rate due to their superior performance. The results are shown in Table 3. It should be noted that we only reported the overall precision, overall recall, and overall F1 of all the methods due to the page limit. Since the class imbalance issue is extremely serious, the overall performance may be dominated by the performance of the dominating class, i.e. the negative samples (reviews without novel need content), which is not the primary concern of the designers, who are more interested in finding the reviews containing novel needs, rather than negative samples. Thus, we also report the recall rate on the positive class in Table 2 shown in the last column.

As the table indicates, the regularized BERT method achieved the best performance in terms of all the performance metrics. The overall precision of all the methods is high, because the overall performance is determined by the negative testing data due to the highly imbalanced class distribution. If we check the class with interest, i.e. the positive sample set, we can see that the recall rate of the proposed method significantly outperforms all others. By contrast, fastText, as a non-deep learning-based approach, could not efficiently extract the semantic information from the text and thus had the worst performance. CNN, LSTM, and transformer-based structure (such as BERT) were the typical methods of encoding the syntactic and semantic information from text. The results of the experiment indicate that BERT was the most efficient in feature extraction and novel need identification.

The proposed regularized BERT structure benefits from adapting the knowledge learned in the pre-trained language models to the specific downstream task. BERT is initially trained on large corpora, capturing general language patterns and nuances (Devlin et al. 2019). By fine-tuning BERT using the annotated reviews on the specific task of identifying novel customer needs, the model leverages its pre-existing knowledge to better adapt to the imbalanced dataset, enhancing its ability to handle both positive and negative reviews.

The proposed regularized BERT method demonstrates clear superiority over established baseline models, such as fastText, CNN, and BiLSTM in the challenging task of identifying novel customer needs from a highly imbalanced online product review dataset. The critical class imbalance issue intensifies the complexity of the task, and regularized BERT effectively addresses this challenge. Unlike traditional models like fastText, which may struggle to capture nuanced patterns in imbalanced datasets and CNN or LSTM that may face challenges in learning dependencies across the sequential nature of reviews, regularized BERT excels due to its ability to capture intricate contextual information and its regularisation mechanisms. The dual BERT networks, driven by the KL divergence regularisation, contribute to a more nuanced understanding of both classes, enhancing the model's sensitivity to minority instances. This leads to improved performance in identifying novel customer needs.

Table 3. Relative performance of BERT and the dual BERT methods.

Testing class	BERT			Regularized dual BERT		
	Overall Precision	Overall Recall	Overall F1	Overall Precision	Overall Recall	Overall F1
Positive class	0.8154	0.7067	0.7571	0.9194	0.7600	0.8321
Negative class	0.9967	0.9982	0.9974	0.9973	0.9992	0.9983
Overall macro average	0.9060	0.8524	0.8773	0.9583	0.8796	0.9152

Table 4. The performance of the proposed method on different testing data lengths.

Testing subsets (number of tokens)	Precision (overall)	Recall (overall)	F1 (overall)
1(1-11)	0.4997	0.5000	0.4999
2(12-27)	0.9219	0.8661	0.8919
3(28-66)	0.9303	0.9400	0.9351
4(66-1632)	0.9976	0.7500	0.8321

The effect of regulation on the dual BERT units

In this section, we show the advantages of the regularized dual BERT-based structure by comparing the performance of identifying reviews with novel need content using the dual BERT-based structure and BERT. The performance on the positive and negative class and the overall performance are reported. The results of the experiment, presented in Table 3, show that the dual BERT structure achieved better overall performance in terms of precision, recall, and F1. The two methods obtained similar performance on the negative class. However, for the positive class, i.e. the identification of reviews containing novel need content, the regularized BERT structure significantly outperformed BERT in all performance metrics.

The dual BERT outperformed BERT, because the regularisation helped to control the model's complexity, preventing it from becoming too specialised in predicting the majority class (Hua et al., 2021). In situations of severe class imbalance, models often struggle to generalise more advantageously to the minority class due to their inherent bias. Regularisation techniques like dropout operations mitigate this issue by encouraging a more generalised and less overfitted model, leading to improved performance in minority class samples.

In addition, deep neural networks are highly prone to overfitting (Dos Santos and Papa 2022). The dropout operation is widely used during model training, which involves randomly setting the values of certain dimensions to 0 in each layer to avoid overfitting. As a result, this leads to randomness in the trained model. In our study, during the identification of positive samples, tested data were also not subjected to any dropout operations. Thus, a discrepancy between model training and testing emerged, which weakened the robustness of the BERT classifier. To improve classification and mitigate randomness, the regularized dual BERT-based network was deployed, such that each text was subjected to two independent BERT methods and yielded two probability distributions across the two classes of reviews. Based on the findings shown in Table 4, outcomes improved as a result. The regularisation of the two outputs thus enhanced not only the model's robustness but also its performance.

Performance of different text lengths

The length of online product reviews exhibited a long-tail distribution, wherein shorter reviews predominated. Statistical analysis revealed that the word count (tokens) in reviews spanned from 4 to 1,632, with a median of 27 words. In this section, we aim to assess the robustness of the trained models concerning varying text lengths.

To conduct our investigation, we partitioned the entire testing dataset into four subsets, each containing an equal number of reviews. The division points, strategically chosen at 11, 27, and 66, align with the first quantile, median, and third quantile of the review lengths. Table 4 illustrates the overall precision, recall, and F1 values for each of these subsets.

The observed pattern in performance improvement across the first three subsets, coupled with a subsequent decline in the longest testing subset, suggests that the model's efficacy in identifying novel needs is influenced by the length of the text. This nuanced analysis sheds light on the model's adaptability and underscores the importance of considering varying text lengths in evaluating its performance.

The observed performance variation in performance across different subsets of testing data segmented by text length can be attributed to the inherent characteristics of reviews with varying lengths and the challenges associated with extracting novel customer needs. The lower performance observed in the subset with the shortest testing reviews can be attributed to the limited context available for the model to discern and understand customer needs effectively. Short reviews may lack the necessary depth and detail for the model to extract meaningful insights. As the length of the testing reviews increases in the first three subsets, the model likely benefits from having more context and information. Longer reviews provide a richer source of data, enabling the model to better capture the nuances and context surrounding novel customer needs.

The decline in performance for the longest testing subset suggests a diminishing return as the length of the reviews continues to increase. This phenomenon aligns with the observation that novel needs are often succinctly expressed in shorter sentences or phrases within the review. Longer reviews may introduce noise and dilute the relevance of key information, making it harder for the model to pinpoint the crucial details. Novel customer needs are frequently expressed concisely, and in longer reviews, essential information may be spread across various sections, making it challenging for the model to identify the specific sentences or phrases that signify novel needs. The dilution effect could lead to a decrease in precision.

The subset with the best performance, which corresponds to Subset 3 with moderate-length reviews, suggests an optimal balance. In this range, the reviews are long enough to provide sufficient context while remaining sufficiently concise for the model to effectively identify and extract novel customer needs.

The effect of training data size

In this section, we investigate the effect of dataset size on the performance of our proposed regularized dual BERT structure. The original dataset consists of 16,667 reviews, with only 187 positive reviews. To study the impact of dataset size, we retrain the dual BERT model using the following data set:

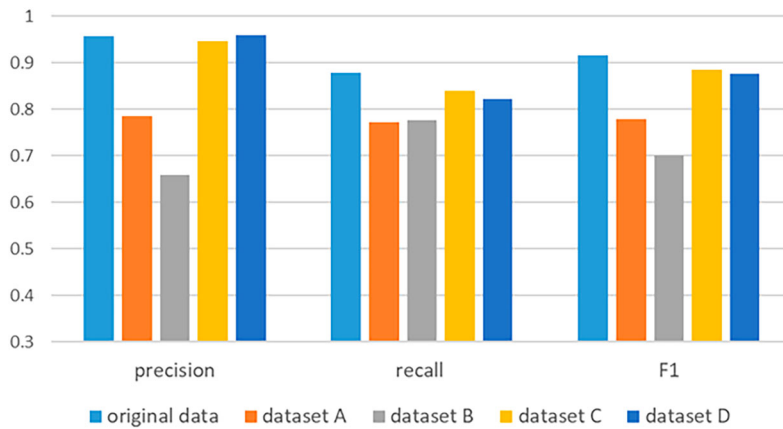


Figure 3. The performance comparison on models training using different datasets.

- Dataset A: Randomly sample half of the whole dataset. This yields similar positive to negative samples ratio.
- Dataset B: Randomly sample one fourth of the whole dataset. This yields similar positive to negative samples ratio.
- Dataset C: Randomly sample half of the negative dataset only and keep the original positive dataset unchanged.
- Dataset D: Randomly sample one fourth of the negative dataset only and keep the original positive dataset unchanged.

The result of these four additional experiments and the original result are shown in Figure 3. The results indicate that the performance of the model drops significantly using dataset A and B for all the three performance measures. In datasets A and B, the overall dataset size is reduced, and the original class imbalance ratio remains the same as they are randomly sampled from the whole original dataset. However, when using dataset C and D to train the model, the performance only drops slightly. The significant drop in performance observed using dataset A and B can be attributed to the reduction in the positive sample size. With fewer positive samples, less information about novel needs content can be extracted, and this leads to worse performance. The model has less data to learn from regarding novel customer needs, which diminishes its ability to generalise and accurately identify these needs.

In contrast, the slight drop in performance in when using dataset C and D can be explained by the fact that the positive sample size remains unchanged. The semantic information extracted regarding novel needs content is not affected much. Only the negative sample size is decreased, but since the original dataset contains a very large number of negative samples, the information on negative samples is still well-covered even with a smaller negative sample size. This allows the model to maintain its performance in identifying novel needs.

Although the overall performance decreases with a smaller dataset, the recall rate does not decrease significantly. In our setting, recall is more important as it represents the model's capability of finding all the reviews containing customer needs. In the product

Table 5. Performance of the model trained using past data.

	Precision	Recall	F1
Positive class	0.8676	0.7468	0.8027
Negative class	0.9967	0.9985	0.9976
Overall macro average	0.9321	0.8727	0.9002

ideation stage, designers can tolerate false positives (i.e. reviews without novel needs content that are misclassified as containing novel needs content) but do not want to have any false negatives (i.e. reviews with novel needs content that are misclassified as not containing novel needs content). Thus, the results show that the model can still effectively identify novel content even with a smaller dataset. This demonstrates the model's robustness and its ability to detect novel customer needs, making it especially useful for product development.

The effect of training data size adaptability to new data

Product reviews are continuously posted online. They serve as an important source for new product development insights. This allows for real-time monitoring of online reviews and the identification of potential product development opportunities. To ensure that the proposed method, which is trained using past data, can effectively function with future data, we conducted an experiment where the model was trained on past data and tested on new data.

We use the data posted online from 2017 to 2021 to train and validate the model and use the reviews posted on 2022 as the testing data. Results can be found in Table 5, which are worse than the results presented in Table 3 but not significantly. It should be noted that the recall of the positive class (i.e. reviews containing novel needs content) is 0.7468, which is slightly worse than the result trained using the original training data, which is 0.7600. As illustrated in the previous section, designers may pay more attention to the recall of the positive class, as this can reveal the model's capability of mining novel needs content from reviews. The results indicate that the model trained on older data (2017-2021) can effectively identify novel customer needs in new reviews (2022). This demonstrates the robustness of our method when applying to new data. This demonstrates that the model can generalise well from past data to future data.

When trained on historical reviews, the model learns to recognise the linguistic structures and sentiment patterns associated with novel needs. The novel needs tend to be hidden within review phrases indicating potential for improvement or additional features that could enhance their experience, such as 'it would be better if the product has ...' or 'the product is good but ...'. The proposed method leverages the power of pre-trained language models like BERT to extract and analyze these sentiments and semantics effectively. By using BERT, the model can understand the context and semantics of the text, allowing it to identify subtle cues and patterns that signify novel customer needs. The regularisation mechanism, which includes KL divergence, helps align the probability distributions generated by the dual BERT networks, ensuring that the model does not overfit to the majority class and maintains a balanced representation of both positive and

negative instances. As a result, it can predict future novel needs that are currently unseen by identifying similar sentiment expressions in new reviews.

Conclusion

In this study, we have developed a regularized BERT structure to identify customer needs with new content from UGC, specifically online product reviews. The importance of identifying novel customer needs from online product reviews is that it has the potential to transform the idea generation process for new product development (Wang et al., 2018). By using deep learning and natural language processing techniques, we provide an effective automated and cost-effective solution to extract valuable insights from UGC. Being able to continuously monitor and identify emerging needs in a timely manner addresses the dynamic nature of consumer demands, giving companies a competitive advantage in responding quickly to market changes. Additionally, our proposed methodology not only simplifies the identification process but also reduces the need for extensive expert resources. This research aligns with the evolving landscape of technology-driven innovation and highlights its potential to reshape industries by empowering designers to extract more creative ideas and convert insights into tangible product opportunities. Ultimately, the outcomes of this project have the potential to advance the current state of product design research and foster a culture of innovation that is adaptable, responsive, and highly attuned to the ever-changing needs of consumers.

Despite its success, our approach has certain limitations. The model's sensitivity to the severity of class imbalance suggests that further research could explore adaptive mechanisms to dynamically adjust regularisation strength based on the data distribution. Additionally, investigating the impact of hyperparameter tuning, such as dropout rates and weight adjustments, could further enhance the model's adaptability (Wu, Wang, and Ma 2021). The generalisation of our findings to other domains and datasets with distinct characteristics warrants exploration. Future work could also focus on extending the regularized BERT approach to multi-label classification scenarios and exploring its efficacy in diverse NLP tasks beyond customer needs identification (Peng et al. 2023). Finally, we would also like to explore how to use transfer learning to adapt the trained model for one particular product to other products.

Data availability statement

The dataset is available at https://anonymous.4open.science/r/Kdaz3HIK4CVFm9GS/novel_generate_v2.xlsx.

Disclosure statement

No potential conflict of interest was reported by the author(s).

Funding

This work was supported by Hong Kong RGC FDS [grant number: UGC/FDS14/E08/23].

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